

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 17:34 UTC

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Generated 2026-05-19 17:34 UTC

- Cycles recorded: **884**
- Time span: 619.9h (~1.43 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	842
BARRIER_HIT	22
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	43
Schur-Weyl Duality	33
Matroid Theory	20
Free Probability Theory	19
Free Probability	18
Representation Theory of Symmetric Groups	18
Additive Combinatorics	14
Noncommutative L^p Geometry	12
Spectral Graph Theory	10
REAL_ALGEBRAIC_GEOMETRY	9
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Finite Geometry	6
Schur-Weyl duality	6
Persistent Homology	6
Noncommutative Harmonic Analysis	6
Plethysm Theory	6
FOURIER_ANALYSIS	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4

Field	Cycles
{ "framework_name": "Ergodic Circuit Framework", "math_branch": "COMM_COMPLEXITY" }	4
Algebraic Topology	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
COMMUNICATION_COMPLEXITY	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic geometry of secant varieties	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-19 13:55 UTC	INCONCLUSIVE	Additive Energy Inverse Proportionality to Karchmer-Wigderson Com
2026-05-19 14:14 UTC	INCONCLUSIVE	Polymatroid Rank Inverse Proportionality to SOS Refutation Size
2026-05-19 14:29 UTC	INCONCLUSIVE	Noncommutative Fourier Coefficient Inverse Proportionality to Com
2026-05-19 14:38 UTC	INCONCLUSIVE	Jordan Rank Lower Bound for Tseitin Resolution Width
2026-05-19 14:50 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportionality to k-CLIQUE Communication Co
2026-05-19 15:23 UTC	INCONCLUSIVE	Real Rank Lower Bound for AC^0 Communication Matrices of PARITY
2026-05-19 15:30 UTC	INCONCLUSIVE	Free Entropy Gap in Read-Twice Branching Programs for Inner Produ
2026-05-19 15:38 UTC	INCONCLUSIVE	Tensor Rank of Clause Incidence Matrix Bounds
2026-05-19 15:42 UTC	BARRIER_HIT	ACC^0 Circuit Size Schur-Weyl Component Count Separation in 3-CNF Incidence Tensors
2026-05-19 15:51 UTC	INCONCLUSIVE	Real Rank of SOS Moment Matrix and Max-CUT Approximation Ratio
2026-05-19 16:12 UTC	INCONCLUSIVE	Persistent Homology Betti Numbers and Communication Complexity of
2026-05-19 16:36 UTC	INCONCLUSIVE	Gröbner Basis Monomial Count and SAT Complexity
2026-05-19 17:04 UTC	INCONCLUSIVE	Free Entropy Lower Bound for Disjointness Communication Complexit

Time	Verdict	Title
2026-05-19 17:21 UTC	INCONCLUSIVE	Schur Positivity of SOS Moment Matrices for 3-SAT
2026-05-19 17:34 UTC	INCONCLUSIVE	Plethysm Coefficient Exponential Gap in Symmetric Powers of Perma

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```